KODURU KARTHIKEYA

Java Backend / Full Stack Software Engineer
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PROFESSIONAL SUMMARY

Final-year BCA student at KL University (CGPA 8.44/10.0) focused on Java backend development with Spring Boot. Built a JWT-secured REST API on PostgreSQL, a React and TypeScript full-stack app, and an AI-assisted Flask app using Gemini.

TECHNICAL SKILLS

Languages: Java, SQL, Python, JavaScript (ES6+), TypeScript
Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT, Maven, Flask, Node.js, Express.js
Databases: PostgreSQL, MySQL, SQLite, MongoDB
Frontend: React, HTML5, CSS3
Testing: JUnit 5, MockMvc
Tools & Deployment: Docker, Git, GitHub, Postman, VS Code, Render, Vercel
Core CS: Data Structures & Algorithms, OOP, DBMS, Java Collections Framework

EDUCATION

Bachelor of Computer Applications (BCA) — KL University, Vaddeswaram, Andhra Pradesh 2024 – 2027 (Expected)
CGPA: 8.44/10.0 | Coursework: DSA, OOP, DBMS, Software Engineering, Web Technologies, AI Fundamentals

EXPERIENCE

Web Development Intern — Thiranex July 2026 – August 2026

- Built responsive user interfaces with HTML5, CSS3, and JavaScript (ES6+).
- Performed cross-browser testing and made layout adjustments for consistent behavior on mobile and desktop.
- Collaborated with the frontend team using Git and GitHub workflows.

Machine Learning Intern — LearnFlu June 2025 – October 2025

- Developed and evaluated classification and regression models in Python on structured datasets.
- Preprocessed data, handling missing values, outliers, and feature scaling.
- Analyzed classification performance using accuracy, precision, and recall.

PROJECTS

NIVOR — Full-Stack Personal Workspace Live Demo | GitHub
Tech: Java 21, Spring Boot, Spring Security, JPA/Hibernate, PostgreSQL, Docker, React 19, TypeScript, Vite, Vercel, Render

- Built a Controller-Service-Repository REST API with 93 endpoint mappings across 20 controllers and 24 JPA entities, using DTOs, Bean Validation, transaction management, and centralized exception handling.
- Implemented stateless JWT authentication with a custom Spring Security filter, BCrypt password hashing, and owner-scoped authorization so users access only their own data.
- Wrote JUnit 5 integration tests with Spring Boot and MockMvc (run on H2 in PostgreSQL mode) covering authentication and cross-user access.
- Containerized the backend with Docker, configured Render deployment with managed PostgreSQL and an /api/health health check, and deployed the React frontend on Vercel.

MediShield — Pharmacy Operations & AI-Assisted Screening Live Demo | GitHub
Tech: Python, Flask, Google Gemini API, SQLite, Gunicorn, Render

- Built a Flask pharmacy operations application with 30+ routes and 8 data models covering medicines, inventory, patients, billing, and PDF invoices.
- Integrated the Google Gemini API (temperature 0) for AI-assisted medicine compatibility screening, using JSON-mode output with server-side parsing, safety validation, and override audit logging.
- Implemented role-based access control (admin, pharmacist, cashier), CSRF token checks, password hashing, and transactional database operations.
- Built billing with 12% GST calculation using Decimal-based rounding, stock deduction and restoration on cancellation, plus CSV/PDF reports, a dashboard, and notifications.

tmfy. — Personalized Digital Notes Web App | *HTML5, CSS3, JavaScript, Vercel* Live Demo

- Built a web app for creating and sharing personalized digital notes via unique links, with one-click clipboard sharing.

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals — June 2026 | AI Agent Builder Masterclass — GSDC
Artificial Intelligence and Machine Learning Courses — Learn Flu (AICTE-approved), 2025

ACHIEVEMENTS

Code4Change 2025: participated in a national-level hackathon at KL University (April 2025), organized by Startup Society.